

4,212,2 Macroeconomics III

Exercises and Independent Studies

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Outline

- ▶ Endogenous Growth Model with R&D (Exercise 5.1a-f)
- ▶ Truly Endogenous Growth Model without Scale Effects (Exercise 5.1g)
- ▶ The Hybrid Model (Exercise 5.2)

Endogenous Growth Model with R&D

$$Y_t = (K_t)^\alpha (A_t L_{Yt})^{1-\alpha},$$

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda,$$

$$K_{t+1} = S_t + (1 - \delta)K_t,$$

$$L_{t+1} = (1 + n)L_t,$$

$$L_{Yt} + L_{At} = L_t,$$

$$L_{At} = s_R L_t,$$

$$C_t + S_t = Y_t,$$

$$S_t = s Y_t.$$

8 Parameters: $\alpha, s, s_R, n, \phi, \rho, \lambda$ and δ . 8 Endogenous variables: Y, K, L, L_A, L_Y, S, C , and A , of which K_t, A_t and L_t are state (predetermined) variables.

Simulations for the Semi-Endogenous Growth ($\phi < 1$)

- ▶ Calibration: $\alpha = 0.33$, $\phi = 0.4$, $\lambda = 0.8$, $\delta = 0.1$, $s = 0.25$, $s_R = 0.04$, $\rho = 1$, $n = 0.04$, $L_0 = 1$.
- ▶ Until period 10 the economy is in steady state. Between period 10 and 11, n drops permanently to $n' = 0.02$.

Endogenous Growth Model with R&D

$$g_{t+1} = (1 + g_t)^{\phi-1} g_t (1 + n)^\lambda$$

$$\tilde{k}_{t+1} = \frac{s\tilde{k}_t^\alpha (1 - s_R)^{1-\alpha} + (1 - \delta)\tilde{k}_t}{(1 + n)(1 + g_t)}$$

$$\tilde{y}_t = \tilde{k}_t^\alpha (1 - s_R)^{1-\alpha}$$

$$k_t = \tilde{k}_t A_t$$

$$y_t = \tilde{y}_t A_t$$

$$g_t^y = \ln(y_t) - \ln(y_{t-1}) = \ln(\tilde{y}_t) - \ln(\tilde{y}_{t-1}) + g_t$$

Endogenous Growth Model with R&D

- ▶ Lower growth rate of the total population means that the growth rate of the labour input in the R&D sector is also lower, implying convergence towards a smaller rate of technological growth in steady state.
- ▶ Lower growth of the labor force implies there is a smaller group of people who will share the capital accumulated in the past. Other things equal, this implies that $\tilde{k}_t$ and $\tilde{y}_t$ increase towards a new and higher steady state value.
- ▶ In the short run, the transitory positive growth of $\tilde{y}_t$ drives the positive contribution of the drop in n to g_t^y , but as the economy approaches the new steady state, g_t^y converges to the new steady state level of g_t .

Initial Value A_0

$$g_{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1$$

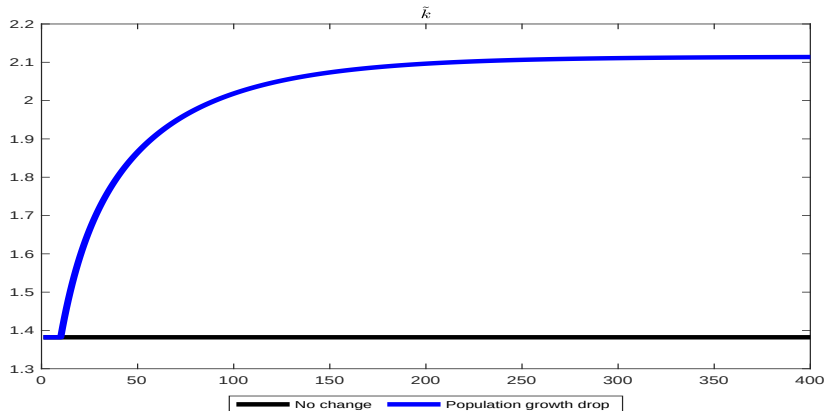
On the balanced growth path the growth rate of technology is constant, $\rho A_t^{\phi-1} L_{At}^\lambda = g_{se}$. This implies that:

$$A_0 = \left(\frac{\rho(s_R L_0)^\lambda}{g_{se}^j} \right)^{\frac{1}{1-\phi}} = \left(\frac{\rho(s_R L_0)^\lambda}{(1 + n)^{\frac{\lambda}{1-\phi}} - 1} \right)^{\frac{1}{1-\phi}} = 1.7905$$

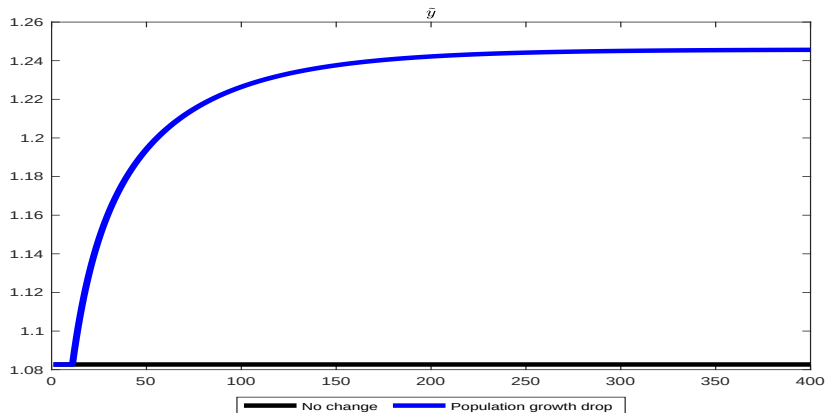
$$g_{se}^j = (1 + n)^{\frac{\lambda}{1-\phi}} - 1 = 0.0537$$

$$g_{se}^f = (1 + n')^{\frac{\lambda}{1-\phi}} - 1 = 0.0268$$

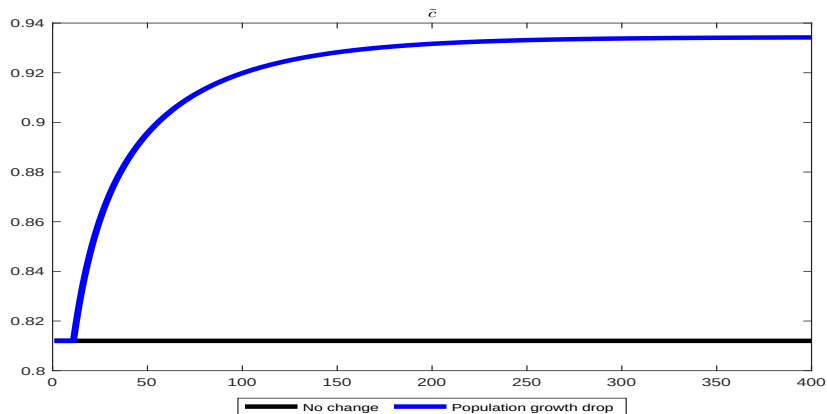
Decrease in the Population Growth Rate



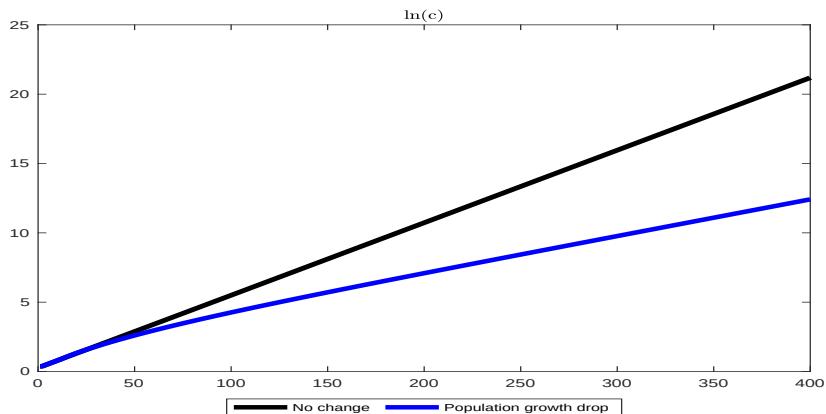
Decrease in the Population Growth Rate



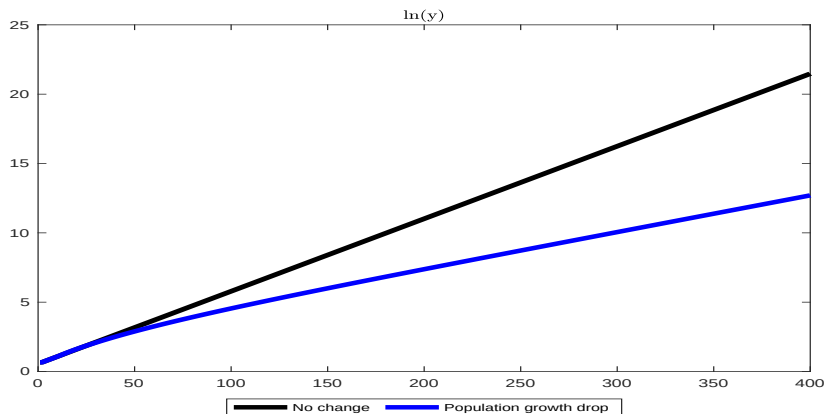
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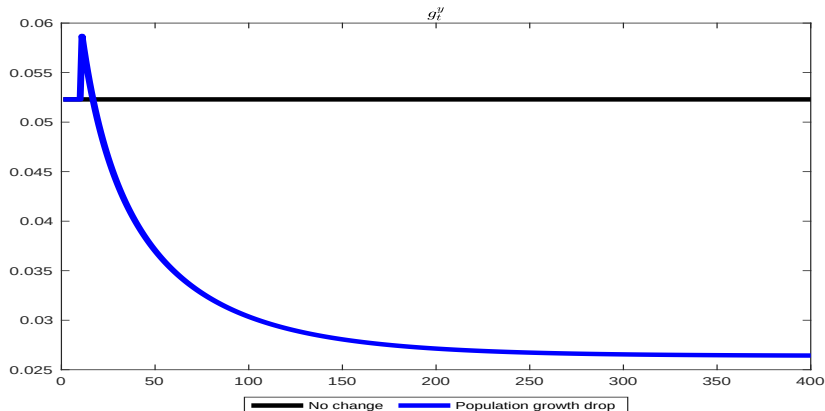
Decrease in the Population Growth Rate



Decrease in the Population Growth Rate

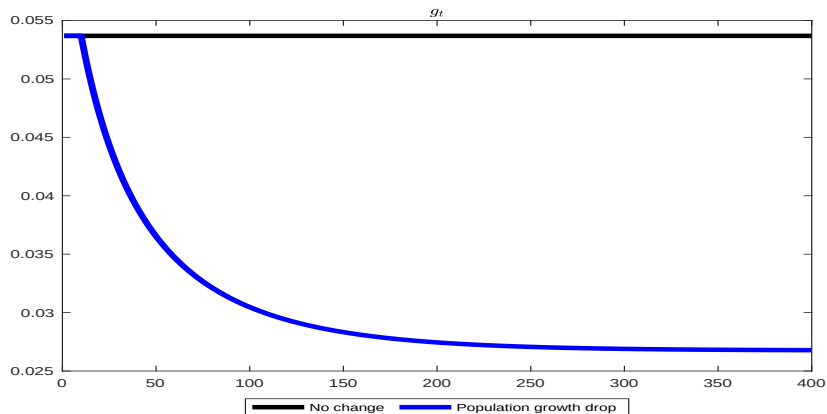


Decrease in the Population Growth Rate

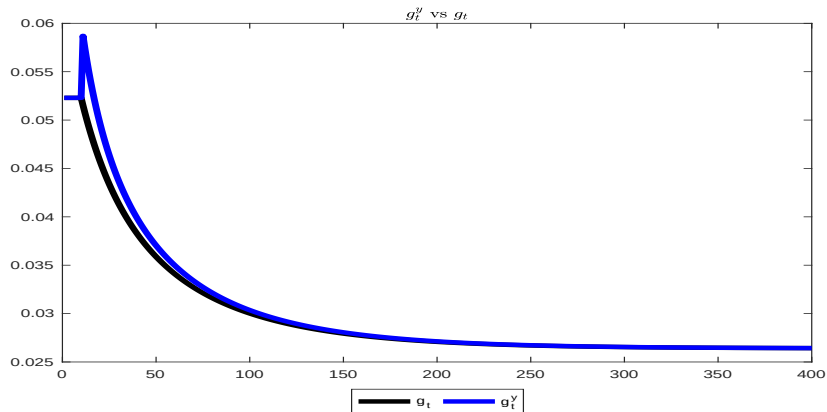


The growth rate of output per worker is above its old steady state value for about 6 periods, so the positive effect on g_t^y from the drop in n is relatively short-lived.

Decrease in the Population Growth Rate



Decrease in the Population Growth Rate



From “Endogenous” to “Truly Endogenous” Growth

- ▶ Models of Chapters 8 and 9:
 - ▶ semi-endogenous: policy could not affect long-run growth (there is positive correlation between output growth and population growth)
 - ▶ truly endogenous with scale effects: larger population implies higher long-run growth
- ▶ Lecture 5a: truly endogenous R&D-driven growth model without scale effect
 - ▶ policy can affect long-run growth
 - ▶ larger population does not imply higher long-run growth (dilution effect eliminates the scale effect): R&D employment as a fraction of the workforce and not as level

“Truly Endogenous” Growth ($\phi = 1$) without Scale Effect

Instead of:

$$A_{t+1} - A_t = \rho A_t L_A^\lambda$$

use:

$$A_{t+1} - A_t = \rho A_t s_R^\lambda$$

This implies that:

$$g_e = \rho s_R^\lambda$$

“Truly Endogenous” Growth ($\phi = 1$) without Scale Effect

$$Y_t = (K_t)^\alpha (A_t L_{Yt})^{1-\alpha},$$

$$A_{t+1} - A_t = \rho A_t s_R^\lambda,$$

$$K_{t+1} = S_t + (1 - \delta)K_t,$$

$$L_{t+1} = (1 + n)L_t,$$

$$L_{Yt} + L_{At} = L_t,$$

$$L_{At} = s_R L_t,$$

$$C_t + S_t = Y_t,$$

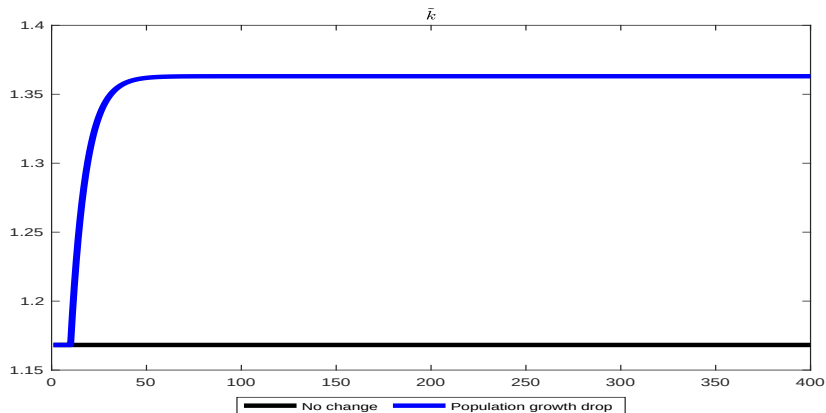
$$S_t = s Y_t.$$

7 Parameters: $\alpha, s, s_R, n, \rho, \lambda$ and δ . 8 Endogenous variables: Y, K, L, L_A, L_Y, S, C , and A , of which K_t, A_t and L_t are state (predetermined) variables.

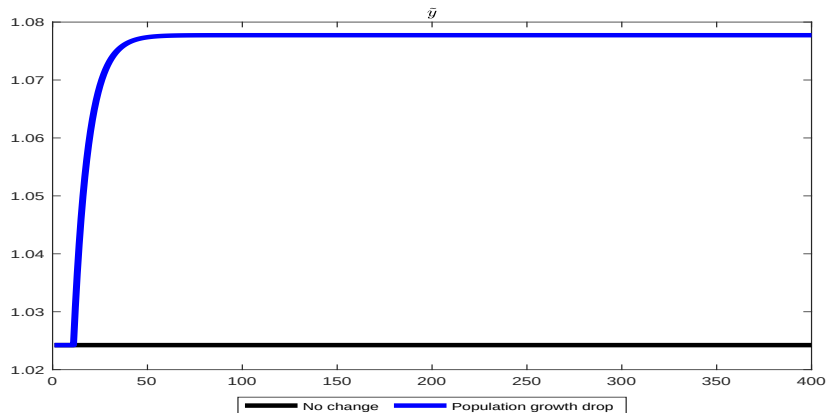
Simulations for the ‘Truly Endogenous’ Growth ($\phi = 1$) without Scale Effect

- ▶ Calibration: $\alpha = 0.33$, $\lambda = 0.8$, $\delta = 0.1$, $s = 0.25$, $s_R = 0.04$, $\rho = 1$, $n = 0.04$, $L_0 = 1$.
- ▶ $\phi = 1$, $A_0 = 1.7905$ (as in the previous exercise) or you may set it equal to 1.
- ▶ Until period 10 the economy is in steady state. Between period 10 and 11, n drops permanently to $n' = 0.02$.

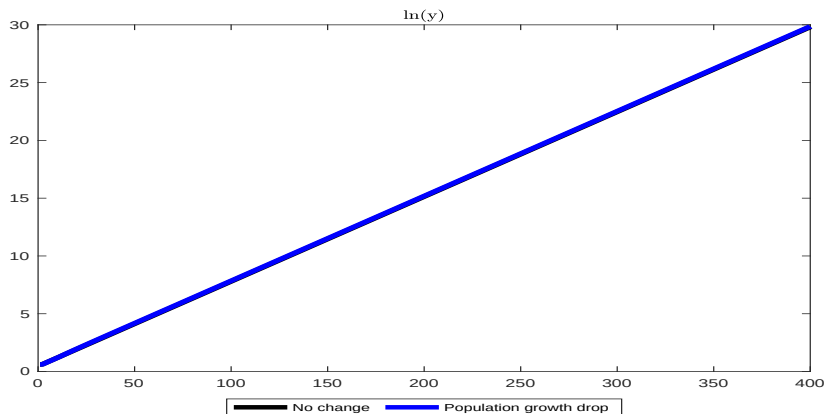
Decrease in the Population Growth Rate



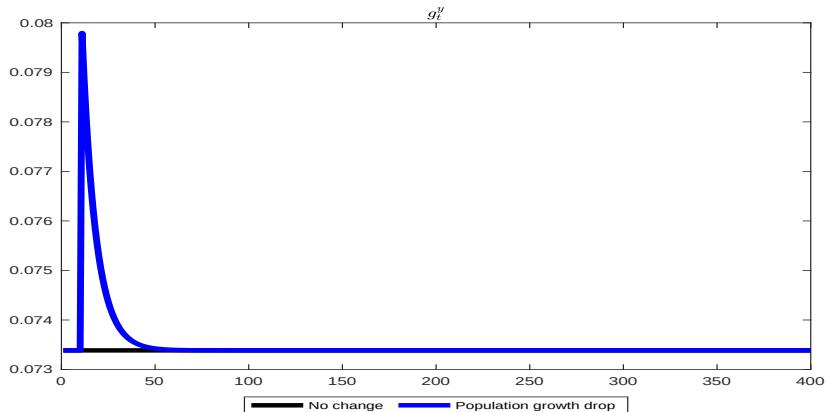
Decrease in the Population Growth Rate



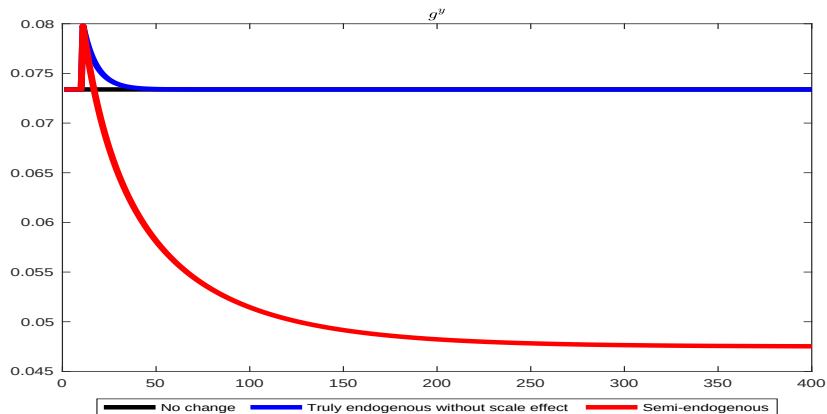
Decrease in the Population Growth Rate



Decrease in the Population Growth Rate



Decrease in the Population Growth Rate



“Truly Endogenous” Growth ($\phi = 1$) without Scale Effect

Take-away of Lecture 5a:

- ▶ Notice that, unlike semi-endogenous growth theory, in this new truly endogenous growth theory without scale effect percapita GDP can be increasing also in the absence of population growth, ($g_e = \rho s_R^\lambda$), that is when $n = 0$, or even with $n < 0$ (shrinking population).
- ▶ Therefore, according to this theory, the long-run growth rate of GDP can persistently be influenced by anything that can affect the fraction of the labour force employed in R&D activities, such as R&D subsidies by the government stimulating private R&D investment and employment.
- ▶ Hence, semi-endogenous models predict that policy can affect growth only temporarily (in the transition), while truly endogenous models predict that policy can affect growth permanently.
- ▶ Which one is right? Estimates of ϕ are close to one, which does not exclude that either of the theories is right!

Cozzi's Hybrid Model

The model assumes that technological progress is a linear combination of the semi-endogenous and the truly endogenous (without scale effect) mechanisms:

$$A_{t+1} - A_t = \alpha_{sem} \rho A_t^\phi L_{At}^\lambda + (1 - \alpha_{sem}) \rho A_t s_{Rt}^\lambda,$$

where α_{sem} is the weight of the semi-endogenous mechanism. This implies that the growth rate of technology is:

$$g_t = \frac{A_{t+1} - A_t}{A_t} = \alpha_{sem} \rho A_t^{\phi-1} L_{At}^\lambda + (1 - \alpha_{sem}) \rho s_{Rt}^\lambda$$

Further assume constant $s_{Rt} = s_R$.

Cozzi's Hybrid Model

$$Y_t = (K_t)^\alpha (A_t L_{Yt})^{1-\alpha},$$

$$A_{t+1} - A_t = \alpha_{sem} \rho A_t^\phi L_{At}^\lambda + (1 - \alpha_{sem}) \rho A_t s_R^\lambda,$$

$$K_{t+1} = S_t + (1 - \delta) K_t,$$

$$L_{t+1} = (1 + n) L_t,$$

$$L_{Yt} + L_{At} = L_t,$$

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9 Parameters: $\alpha, \alpha_{sem}, s, s_R, n, \phi, \rho, \lambda$ and δ . 8 Endogenous variables: Y, K, L, L_A, L_Y, S, C , and A , of which K_t, A_t and L_t are state (predetermined) variables.

Implications

Only the solution for technological growth needs to be discussed as the rest can be solved as in the model of Chapter 9.

- Constant population $n = 0$, $L_t = L$ leads to:

$$g_t \xrightarrow{t \rightarrow \infty} (1 - \alpha_{sem}) \rho s_R^\lambda = (1 - \alpha_{sem}) g_e$$

- Shrinking population $n < 0$, $L_t \xrightarrow{t \rightarrow \infty} 0$ leads to:

$$g_t \xrightarrow{t \rightarrow \infty} (1 - \alpha_{sem}) g_e$$

Implications

- Increasing population $n > 0$, $L_t \xrightarrow{t \rightarrow \infty} \infty$ leads to:

$$\frac{g_{t+1} - (1 - \alpha_{sem})\rho s_R^\lambda}{g_t - (1 - \alpha_{sem})\rho s_R^\lambda} = (g_t + 1)^{\phi-1}(1 + n)\lambda$$

In the steady state, $g_{t+1} = g_t = g^h$:

$$g^h = (1 + n)^{\frac{\lambda}{1-\phi}} - 1 \equiv g_{se}$$

- This result holds only if $g_{se} > (1 - \alpha_{sem})g_e$. Otherwise, the steady state growth would equal $(1 - \alpha_{sem})g_e$.

Implications

- ▶ This result holds only if $g_{se} > (1 - \alpha_{sem})g_e$. Otherwise, the steady state growth would equal $(1 - \alpha_{sem})g_e$.
- ▶ Why?
- ▶ We know that in the long run, the economy will grow at rate $(1 - \alpha_{sem})g_e$ if the first term of $\alpha_{sem}\rho A_t^{\phi-1} L_{At}^\lambda + (1 - \alpha_{sem})\rho s_{Rt}^\lambda$ will converge to zero.
- ▶ When that could happen for $n > 0$?
- ▶ Only if $A_t^{\phi-1}$ grows faster than L_{At}^λ , or equivalently (as written in the slides) when A_t grows faster than $L_{At}^{\frac{\lambda}{1-\phi}}$. Therefore, in order to have the long run (steady state) growth rate equal to g_{se} it is not sufficient to have $n > 0$ but that $n > \bar{n} > 0$.

Threshold Level $\bar{n}$

- ▶ The semi-endogenous growth rate g_{se} will prevail if and only if the population growth rate, n , is higher than a threshold level, $\bar{n}$:

$$n > \bar{n} \equiv \left[(1 - \alpha_{sem}) \rho s_R^\lambda + 1 \right]^{\frac{1-\phi}{\lambda}} - 1$$

- ▶ In the long-run, per-capita GDP will follow the predictions of the semi-endogenous growth theory if population growth rate n is larger than a positive threshold $\bar{n}$.
- ▶ In the long-run, per-capita GDP will follow the predictions of the fully endogenous growth theory without scale effects if population growth rate n is lower than $\bar{n}$.

Calibration of Exercise 5.2

- ▶ The parameters for *Economy 1* are given by $\alpha = 0.33$, $\alpha_{sem} = 0.4$, $\phi = 0.4$, $\lambda = 0.8$, $\delta = 0.1$, $s = 0.25$, $s_R = 0.04$, $\rho = 1$, $n = 0.04$, $L_0 = 1$, $K_0 = 1$, $A_0 = 1$.
- ▶ *Economy 2* is characterized by the same parameters except for the population growth rate, which happens to be $n = 0.02$.

Steady State

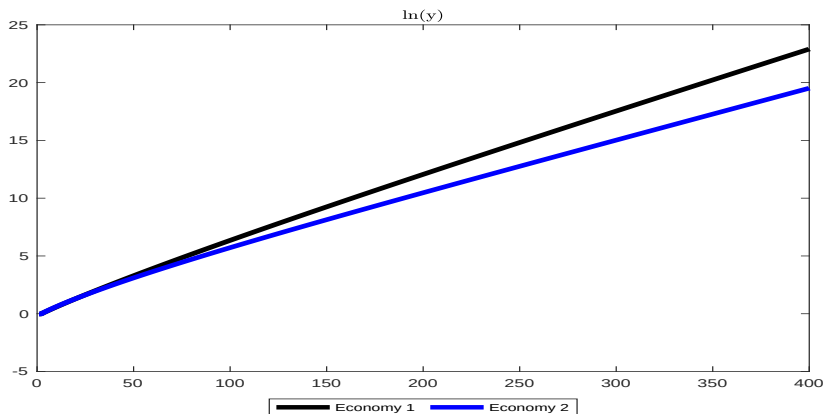
$$\begin{aligned}\bar{n} &= \left[(1 - \alpha_{sem}) \rho s_R^\lambda + 1 \right]^{\frac{1-\phi}{\lambda}} - 1 \\ &= 0.034\end{aligned}$$

- ▶ *Economy 1*: $n = 0.04$, it implies that $g^h = g_{se} = 0.054$. Growth is semi-endogenous on the balanced growth path.
- ▶ *Economy 2*: $n = 0.02$, it implies that $g^h = (1 - \alpha_{sem})g_e = 0.046$. Growth is endogenous on the balanced growth path.

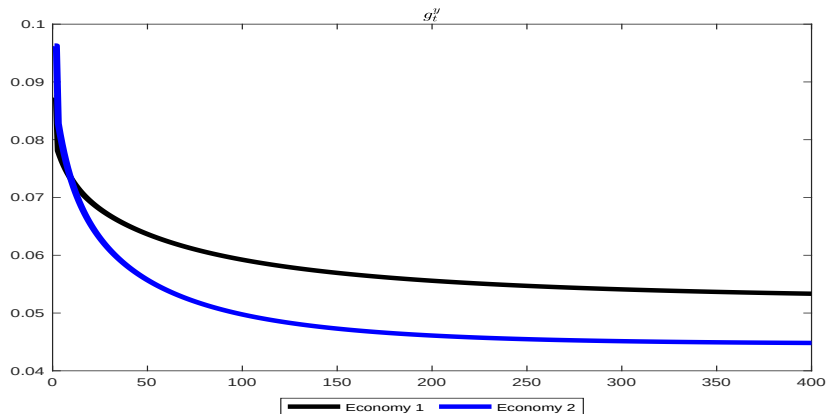
Table: Comparison of Economies

	Economy 1 (n=0.04)	Economy 2 (n=0.02)
g	0.054	0.046
$\tilde{k}^*$	1.382	1.759
$\tilde{y}^*$	1.083	1.172
$\tilde{c}^*$	0.812	0.879

The evolution of $\ln(y_t)$



The evolution of g_t^y



The evolution of g_t^y and g_t

